

RESEARCH PROBLEM

Every hour, someone decides how much electricity Ontario's power plants should produce — and today that decision **completely ignores the carbon cost of burning gas**. With Canada raising its carbon price to **\$170/tonne by 2030**, Ontario's grid urgently needs a smarter tool: one that can look at any given day, forecast what demand will be, and automatically figure out **the cheapest way to supply that electricity while producing as little CO₂ as possible**. That tool is **OMEGA**.

96K

hourly records
2015–2025

46

engineered
features

750MW

avg gas dispatch
peaks 5,189 MW

67.2M

tonnes CO₂
10-year study

2,291 hrs exceeded 100 g CO₂/kWh · Renewables avg only 34% · No predictive dispatch exists

RESEARCH QUESTIONS

- **RQ1:** Is there a specific carbon price that triggers Ontario's grid to switch away from gas — and what is it?
- **RQ2:** If carbon pricing forces cleaner dispatch, how much extra does it cost the average Ontario household per month?
- **RQ3:** Can combining multiple ML models predict Ontario's electricity demand better than any single model alone?

HYPOTHESES & OUTCOMES

H1 ✓ Ridge matches/outperforms all base models — auto-specialisation confirmed
LSTM leads solar (+0.905★), GBM leads demand (+0.608★) · Ridge R²=0.832 vs GBM=0.830

H2 ✓ Non-linear step-change at \$80/t · ≥10% CO₂ reduction confirmed vs \$0 baseline
Gas: 38,730 → 26,850 MWh (–31%) · CO₂: –5,821t (–13%) · \$0/\$50/\$72 identical

H3 ✓ Household cost stays under \$1.00/month even at \$250/tonne
Only +\$0.56/mo per household — 900 kWh/mo × \$0.617/MWh extra

DATA & SOURCES

96,264 hourly records · Jan 2015–Dec 2025 · 3 forecast targets (demand, solar, wind)

IESO Open Data · NASA POWER · ECCC NIR · Our World in Data

Split: 90% train (Jan 2015–Aug 2024) · 5% val · 5% test — chronological, no shuffle

Features: temporal · lag (1h/24h/168h) · rolling 24h stats · weather · season encoding · 3-layer analog proxy

Gas mean 750 MW vs peak 5,189 MW — 7× variability is why Ontario needs a predictive dispatch optimizer, not reactive market rules

TECHNOLOGY STACK

Python 3.11 · scikit-learn · TensorFlow/Keras · PuLP + CBC · Streamlit · Power BI

GCP Vertex AI

ANALYTICAL PIPELINE — TWO-MODULE SYSTEM

